Market and Patent Research Report.

ProdDev SS25 - Group A

April 14, 2025

1 Introduction

Invasive alien plant species such as Heracleum mantegazzianum, Lupinus polyphyllus, and Jacobaea vulgaris pose significant environmental and public health challenges in Europe. These plants negatively affect native biodiversity, outcompete local flora, and may pose toxicological risks to humans and animals. Conventional removal methods rely on herbicides, which are increasingly restricted by EU environmental legislation (e.g., Regulation (EU) No. 1143/2014), or labor-intensive mechanical clearing. Recent advances in agricultural robotics provide an opportunity to address this issue using autonomous systems that employ machine learning, vision technologies, and non-chemical removal techniques, including electric pulse applications.

2 Market Overview and Technological Landscape

2.1 RootWave (UK) – Electric Weed Control System

- **Technology:** Applies high-frequency electric pulses through contact probes to thermally destroy plant cells from root to tip.
- **Target Species:** Effective on deep-rooted perennials including *Heracleum mantegazzianum.*
- **Advantages:**
  - Non-chemical and environmentally safe.
  - Minimal soil disturbance.
  - Suitable for robotic integration. Carbon Robotics
- **Use Cases:** Municipalities, ecological reserves, organic agriculture.

Figure 1: Source: RootWave

Video Demonstration:

- RootWave Demonstration 1
- RootWave Demonstration 2

2.2 Carbon Robotics – LaserWeeder

- **Technology:** AI-powered machine vision with mounted CO_2 lasers for precision weed eradication.
- **Target Species:** More suitable for row crops. Not designed specifically for tall or toxic plants like hogweed.

Figure 2: Source: RootWave

- **Video Demonstration:** LaserWeeder Overview

2.3 Odd. Bot – Weed Pulling Robot (NL)

- **Technology:** Selectively detects and pulls weeds using robotic grippers, based on image classification algorithms.
- **Target Species:** Applicable for broadleaf weed species like *Jacobaea vulgaris*.

Figure 3: Source: RootWave

2.4 Drone and Aerial Detection Platforms (DK, DE, UK)

Developer: University of Connecticut (BirdHabitatBot)

- **Technology:** Drones combined with NDVI, hyperspectral imaging, or machine learning can map infestations of hogweed in open fields and forests.
- **Advantages:** Ideal for pre-mapping and targeting robotic ground intervention.

Figure 4: Source: RootWave

2.5 FarmWise – Titan FT-35 (USA)

- **Manufacturer:** FarmWise
- **Technology:** A self-driving robotic cultivator with vision and mechanical blade system for precision weeding. Optional fertilizer or pesticide application.
- **Target Species:** Broadleaf weeds in vegetable crops.
- **Use Cases:** Commercial vegetable farms.

2.6 Naïo Technologies – Dino (France)

- **Manufacturer:** Naïo Technologies
- **Technology:** GPS navigation, laser scanners, vision-based selective mechanical weeding.
- **Target Species:** Weeds between and near crop rows.
- **Use Cases:** Organic farms requiring non-chemical weeding.

2.7 ecoRobotix – AVO (Switzerland)

- **Manufacturer:** ecoRobotix
- **Technology:** Solar-powered robot with precision herbicide spray using machine learning.
- **Target Species:** Isolated weeds among broadleaf crops.
- **Use Cases:** Sustainable farming.

2.8 FarmDroid – FD20 (Denmark)

- **Manufacturer:** FarmDroid
- **Technology:** GPS-based seeding and blind mechanical weeding.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** Organic grain and vegetable farms.

2.9 Small Robot Company – "Tom & Dick" (UK)

- **Manufacturer:** Small Robot Company
- **Technology:** Tom scouts and maps, Dick applies electric shock to kill weeds.
- **Target Species:** Broadleaf weeds in high-value crops.
- **Use Cases:** Precision agriculture.

2.10 AutoWeed – Lantana Targeting Robot (Australia)

- **Developer:** James Cook University
- **Technology:** AI and cameras detect and spray herbicide on *Lantana camara*.
- **Target Species:** *Lantana camara*.
- **Use Cases:** Biodiversity protection.

2.11 BirdHabitatBot – Forest Invasives Robot (USA)

- **Developer:** University of Connecticut
- **Technology:** Drone-GPS mapping + ground robot to remove invasive shrubs.
- **Target Species:** *Japanese barberry*, *multiflora rose*.
- **Use Cases:** Forest conservation.

2.12 Bosch Deepfield – BoniRob (Germany)

- **Developer:** Bosch Research
- **Technology:** Vision-guided pneumatic "stamping" system.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** High-speed non-chemical cultivation.

2.13 Hydrilla Hunter – Robotic Boat (USA)

- **Developer:** Northeastern University & CAES
- **Technology:** Hyperspectral aquatic detection and robotic navigation.
- **Target Species:** *Hydrilla*, *Eurasian watermilfoil*.
- **Use Cases:** Freshwater habitat protection.

3 Patent Research Overview

3.1 EP3804483A1 – Tool for Eradication of Invasive Hogweed Species

- **Assignee:** Søren Lauritzen (Denmark)
- **Description:** Manual stem-puncturing tool to induce necrosis in *Heracleum mantegazzianum*.
- **Significance:** Basis for robotic puncturing adaptation.
- **Link:** EP3804483A1

3.2 WO2021055485A1 – Autonomous Laser Weed Eradication System

- **Assignee:** Carbon Robotics
- **Description:** Robotic platform guiding laser source for weed removal.
- **Limitation:** Not optimized for toxic plants or non-row fields.
- **Link:** WO2021055485A1

3.3 US20170238460A1 – Weeding Robot and Method

- **Assignee:** Blue River Technology
- **Description:** Selective weeding robot using sensors and mechanical tools.
- **Potential Use:** Adaptable for invasive species.
- **Link:** US20170238460A1

3.4 EP3744173A1 – Weed Inactivation Device Using Electric Pulses

- **Assignee:** Zasso GmbH
- **Description:** Electrode-based high-voltage weed inactivation.
- **Relevance:** Integrable into autonomous platforms.
- **Link:** EP3744173A1

3.5 DE102015209879A1 – Device for Damaging Weeds

- **Assignee:** Bosch GmbH
- **Description:** Tool with classification and positioning unit for targeted weed damage.
- **Significance:** Precise, rapid chemical-free weed control.
- **Link:** DE102015209879A1

3.6 DE4039797A1 – Weed Control Device

- **Assignee:** Robert Bosch GmbH
- **Description:** Continuous weed destruction system interrupted only when crop detected.
- **Significance:** Efficient for automated systems.
- **Link:** DE4039797A1

Traide Fair Australia

During a recent trade fair visit (14.04.2025) to XAG Sydney Australia, supported by further online research, the **XAG P150 agricultural drone** was identified as a promising solution for the removal of invasive plant species. The drone features a payload capacity of up to 70 kg and uses the **RevoSpray 4 system** to apply herbicides either across wide areas or with pinpoint accuracy. The spraying system delivers fine droplets via adjustable centrifugal nozzles, enabling effective coverage on both sides of plant leaves.

The process is managed using **PIX4Dfields**, a mapping and control software. First, a drone equipped with a high-resolution camera captures detailed images of the area. In the software, the user manually marks invasive species, protected plants, and obstacles. These inputs train the software to recognize similar vegetation patterns using machine learning. The generated prescription map is then uploaded to the spraying drone, which treats only the marked zones with high precision.

Compared to ground-based mowers like the **ROBO MOWER T-90** by NetEvoTech, spraying drones offer several advantages. Mechanical mowing can lead to seed dispersion and is limited in difficult or uneven terrain. Experts advised against the use of mowers in this context and recommended drone application to avoid further spread and ensure better reach. The system is also compatible with drone charging stations from **Enovance**, which can quickly recharge batteries for continuous operation. The total cost for a two-drone system including software is approximately **AUD 50,000**.

Sources:

XAG P150 Product Page: https://www.xa.com/en/p150

PIX4Dfields Software Overview: https://www.pix4d.com/product/pix4dfields

RevoSpray 4 Spraying System: https://www.xa.com/en/revospray

ROBO MOWER T-90 Specs (NetEvoTech): https://www.netevotech.com/robo-mower-t90

Enovance Drone Charging: https://enovance.com.au/


XAG
XAG® P150 Agricultural Drone
Better Coverage, Healthier Crops, Bigger Yields!
Farming isn't just about working harder — it's about working smarter.
Flight Speed
18 m/s[1]
Payload
70 Kg[2]
Max. Flow Rate
30 L/min
Max. Spread Rate
280 Kg/min[3]
Real-Time 3D Terrain Mapping
Variable Rate Application
1. Maximum Payload: The maximum payload may differ depending on the task system and is influenced by operating conditions and parameters.
2. Flight Speed Regulations: The maximum flight speed is subject to local laws and regulations. Performance may vary based on environmental factors, temperature and user operation. Specific conditions require optimized parameters, significantly impacting efficiency.
3. Maximum Spread Rate: The maximum spread rate of 280 kg/min is measured with a 50 kg bag of compound fertilizer while 220 kg/min is measured with urea. Spread rates vary based on fertilizer granule size, density and surface smoothness.
XAG-AU.COM


Take a Closer Look
Collapsible, Low-Maintenance and Modular Agricultural Drone
4D Imaging Radar
Detect and bypass obstacles within 1.5-100m, adapts to varied terrain autonomously
SuperX 5 Pro Intelligent Control System
10x increase in computing power, fly smarter and perform better
Foldable Arms
Flexible to transfer
Robust Body Structure
Easier to maintain and cost-saving
Ready When You Are
Deploy anytime, anywhere — no scheduling conflictsor downtime.
Take control of your operations with on-demand deployment.
Say goodbye to contractor delays and tight schedules — your drone is always ready when you are.
Built for Tough Terrain
Consistent performance in wet paddocks, hillyterrain, or small fields.
Engineered to handle Australia's diverse landscapes this drone delivers reliable results where ground rigs struggle — no matter the conditions.
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XAG-AU.COM
Application Scenarios
Spraying Parameters
30 L/ha
Dosage Rate
9 Meters
Spray Width
18 m/s
Flight Speed
Field-Proven Spray Efficiency
26 Hectares Per Hour
Spreading Parameters
300 Kg/ha
Discharge Rate
6.6 Meters
Spread Width
18 m/s
Flight Speed
Field-Proven Spread Efficiency
2,167 Kg of granules Per Hour
Spraying
with XAG® P150 RevoSpray P4
Revolutionise crop protection with intelligent rotary atomization technology designed for superior efficiency and uniform coverage.
5-10 m
Spraying Width
70 L
Smart Liquid Tank
60-500 μm
Droplet Size Range
Spreading
with XAG® P150 RevoCast P4
Cover more ground with superior control, efficiency, and versatility.
280 Kg/min
Max. Spread Rate
115 L
Granule Container
3-8 m
Effective Spread Width
Field Mapping
with XAG® P150 RealTerra
Revolutionise your farming with real-time 3D mapping technology.
Up to 13.3 Hectares
Survey area in a single flight
HD maps & 3D modeling
Types of maps
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XAG® P150 Agricultural Drone

XAG

- Delivers precise spraying and spreading, ensuring crops get exactly what they need.
- Reduce overspray, save on chemicals, and maximize crop health.
- Works when you need it — no waiting for pilots or weather windows.

XAG® SRC4 Remote Controller

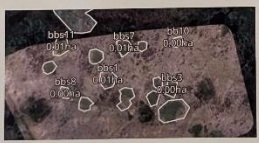
Elevate your agricultural drone operations with 6-inch ultra-bright screen, 2 km range, 5-hour battery life, and real-time connectivity for precision agriculture.

XAG® DL1 Data Link

Designed to extend your drone's communication range, eliminate signal dead zones, and ensure uninterrupted operations in even the most challenging environments.

XAG® XRTK6 Pro Mobile Station

Achieve centimetre-level accuracy with the XAG® XRTK6 Pro Mobile Station. Portable RTK, 1.5 km Wi-Fi range, and offline satellite correction for precision farming.



Spot Spraying

Multi-Field Application — Ideal for small fields or targeted area applications.

Swarm Control

Swarm Operation allows for the autonomous control of two drones simultaneously.

XAG® Care P150

Fly with Peace of Mind

Comprehensive coverage for your XAG® P150, so unexpected repair costs don't slow down your farming success.

Visit Our Website for More Information

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NetEvoTech
ROBO MOWER T-PREMIUM
Stronger Build with Eco Power Option
Battery Version Available
Choose the battery-powered option of T-PREMIUM for lower noise levels and eco-friendly, emission-free performance.
Reinforced Tracks
Upgraded Dual Blades
Powerful Lift System
NetEvo Tech
Debris Protection
Optional Accessories
Snow Plough
Rotary Brush
Industrial Remote Control
Remote Control with Display and Camera
Tow Hitch
NetEvo Tech
Upgraded Enclosed Design
The advanced enclosed design offers superior protection, enhancing the safety and longevity of both the machine and the operator.
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NetEvoTech

ROBO MOWER T-SERIES

Performance Features

- **Advanced brushless electric motor:** Efficiently powers the tracks, delivering smooth, reliable movement.
- **Durable hardened steel blades:** Built to resist wear, guaranteeing long-lasting sharpness.
- **Long-range remote control:** Manage operations effortlessly from up to 200 meters away.
- **Reinforced rubber tracks:** Engineered for durability on any terrain, giving unmatched stability.
- **Robust, impact-resistant design:** Crafted to minimize damage, ensuring top-tier resilience in all conditions.
- **Self-charging hybrid drive:** Continuously powers up while the mower works, ensuring uninterrupted performance.
- **High-powered gasoline engine:** Fuels the cutting blades with strength for superior mowing efficiency.

CE Certificate EPA Certificate

Optional Automation Add-on

Take your mowing to the next level with the optional automation module, enabling precise, automated navigation along pre-planned routes.

All-Terrain Mastery

Shrub Clearing

The sharp blades and powerful engine of the T-Series ensure clean, easy cuts through small shrubs.

Hillside Protection

Reinforced heavy-duty tracks with and low centre of gravity ensure stability on slopes up to 65°.

Solar Panel Fields

The T-Series' super-low profile and 360° rotation easily navigate under solar panels and other low obstacles.

Orchard Maintenance

Maneuverability allows smooth cutting under trees, without damaging fruit trees or branches.

Embankment Maintenance

The T-Series' powerful climbing ability ensures efficient maintenance of steep embankments.

NetEvoTech

ROBO MOWER T-SERIES

Equipped with a powerful engine, the T-Series ROBO Mowers are **built for the toughest terrains,** including steep slopes. Ideal for **large farms, orchards, starting a business in utility mowing, or heavy-duty commercial use,** it combines power with a wide cutting width to handle any challenge with ease.

Compare Model Specification:

	 T-90	 T-PREMIUM
Dimensions:	140cm x 130cm x 63cm	140cm x 140cm x 70cm
Weight:	315kg	360kg
Mowing width:	90cm	90cm
Efficiency:	3000-4500m²/h	3000-4500m²/h
Engine type:	Four-stroke engine	Four-stroke engine
Speed of movement:	0-5km/h	0-5km/h
Maximum working slope:	65°	65°
Blade lifting range:	1-20cm	1-20cm
Maximum cutting height:	2m	2m
Remote control range:	200m	200m


NetEvoTech
ROBO MOWER T-90
Unleash the Power, Conquer Any Terrain
Reinforced Tracks
Upgraded Blades
Powerful Lift System
NetEvo Tech
Debris Protection
Optional Accessories
Cargo Platform
Atomizer 200L
Horizontal and Vertical Sprinkler 200L
Snow Plough
Rotary Brush
Industrial Remote Control
Remote Control with Display and Camera
Tow Hitch
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enovance
Overall solution for drone charging
Model: O-Power 21 (Battery Pack)
Model: IBC6000W (AC-DC Charger)
Model: IBC10000W (DC-DC Charger)
Efficient & Time-saving
Battery Pack takes only 6 hours to fully charge with 220V mains power, and only 2 hours with the EV DC fast charger.
Only takes about 8 mins to charge the drone battery.
Silent & Zero Pollution
No noise pollution, no maintenance required.
Worry-free & Reliable
3000 cycles warranty, worry-free.
Two-year warranty, free repair for non-man-made faults.
Accidental spontaneous combustion insurance up to 3 million dollars.
Economical & Affordable
90% reduction in charging costs compared to gasoline generators.
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UN38.3

enovance

Datasheet

O-Power 21 (Battery Pack)

Model	O-Power 21 (Battery Pack)
Cell technology	$LiFePO_4$
Dimension (D*W*H)	820×580×295mm
Package size (D*W*H)	1300×730×540mm
Weight(Net weight)	174.5kg
Nominal capacity	280Ah
Nominal energy	21.5kWh
Battery group mode	1P24S
Rated charge/discharge current	140A
Max.Charge/Discharge current	200A
Operating voltage	67.2V - 84.24V
Nominal voltage	76.8V
Recommended crate	0.5C
USB interface voltage	5V
USB interface power	65W
Display	4.3 inches
Heat dissipation method	Air cooling
Ingress protection	IP20
Working altitude	≤3000m
Operating temperature	0-50℃
Cycle life	≥3000 Cycles

IBC6000W (AC-DC Charger)

Model	IBC6000W (AC-DC Charger)
Dimension (D*W*H)	450×223.4×66.2mm
Weight (Net weight)	8.5 kg
Rated input voltage	220Vac
Input voltage range	90Vac-264Vac (90 - 185Vac derating 2000W output)
Output voltage range	60Vdc - 87.6Vdc
Output current	0 - 70A
Power	6kW
Input method	L+N+PE
Heat dissipation method	Air cooling
Ingress protection rating	IP20
Working altitude	≤3000m
Operating temperature	0-50℃

IBC10000W (DC-DC Charger)

Model	IBC10000W (DC-DC Charger)
Dimension (D*W*H)	550×230×94.5mm
Weight (Net weight)	14.5kg
Input voltage range	65Vdc-100Vdc
Output voltage range	0.5Vdc-58.8Vdc
Output current	0 - 180A (Default 150A)
Output power	10kW
Heat dissipation method	Air cooling
Ingress protection	IP20
Working altitude	≤3000m
Operating temperature	0-50℃


M4
SMART PADDOCK


XAG® P150
RevoCast P4
XAG® P150
RevoSpray P4
XAG P150
TRADE-IN DEALS
M2
M1
Certified Drone Training
XAG


XAG® P150
RevoCast P4
XAG
AUSTRALIA
Effortless precision,
unmatched productivity
115 L
Granule Container
280 Kg/min
Max. Spread Rate
3-8 m¹
Spread Swath
70 Kg
Payload
2,167 Kg/hr²
Spreading Efficiency
Versatile Screw Feeders
Vertical Waving Spreader
Dual-Side Large Filling Ports
Operational Influences: Actual operation data is influenced by environmental factors, temperature, and user handling.
Follow official product instructions, as performance may vary under different conditions.
Spreading Efficiency: Measured at a 300 kg/Ha application volume, 6.8 m spacing, and 18 m/s speed in open fields.
Actual performance depends on real-life conditions.
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18 m/s


XAG® P150
RevoSpray P4
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Pioneering intelligent rotary atomisation for superior crop protection in any environment
18 m/s Max. Flight Speed
30 L/min Max. Flow Rate
5-10 m Spraying Width
26 ha/h Spraying Efficiency
60-500 μm Droplet Size Range
70 L Smart Liquid Tank
Advanced Centrifugal Nozzles
Flexible Impeller Pumps
Spraying Efficiency: The spraying efficiency is measured at a 20 L/Ha application volume, 10 m spacing, and 18 m/s speed in open fields. Actual performance may vary due to real-life conditions.
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XAG AUSTRALIA
3-8 m Spread Swath
Kg/hr Efficiency
Dual-Side Large Filling Ports


XAG® P150
XAG
AUSTRALIA
TRADE-IN DEALS AVAILABLE:
UPGRADE
XAG® P150
up to 26 ha/h
OVERALL SPRAYING EFFICIENCY 1
Older Model
up to 19 ha/h
OVERALL SPRAYING EFFICIENCY 2
≈ more than a 36% increase in productivity
Contact us now for a free valuation on your current models trade in!
COMPARISON WITH OLDER AGRICULTURAL DRONES
Tank Size
Max Operational Speed
Max Flow Rate
P150 70L 18 m/s 30 l/min
P100 Pro 50L 13.8 m/s 22 l/min
P100 40L 13.8 m/s 12 l/min
V40 16L 10 m/s 10 l/min
P30 16L 8 m/s 5.6 l/min
Other 40L 10 m/s 12 l/min
Spraying Efficiency: The spraying efficiency is measured at a 20 L/Ha application volume, 10 m spacing, and 18 m/s speed in open fields. Actual performance may vary due to real-life conditions.
19 ha/h is for normal large-volume spraying (such as wheat pest control), P100 Pro operating parameters: 30 L/ha, 9 m flight route spacing, 13.8 m/s flight speed.
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Spraying Efficiency
Droplet Size Range
70 L Smart Liquid Tank
Advanced Centrifugal Nozzles
Flexible Impeller Pumps
Versatile Screw Feeders
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XAG
AGRICULTURAL DRONES
Tank Size | Max Operational Speed | Max Flow Rate
P150 | 70L | 18 m/s | 30 l/min
P100 Pro | 50L | 13.8 m/s | 22 l/min
P100 | 40L | 13.8 m/s | 12 l/min
V40 | 16L | 10 m/s | 10 l/min
P30 | 16L | 8 m/s | 5.6 l/min
Other | 40L | 10 m/s | 12 l/min

115 L
Granule Container
70 Kg
Payload
Versatile Screw Feeders
Vertical Waving Spreader
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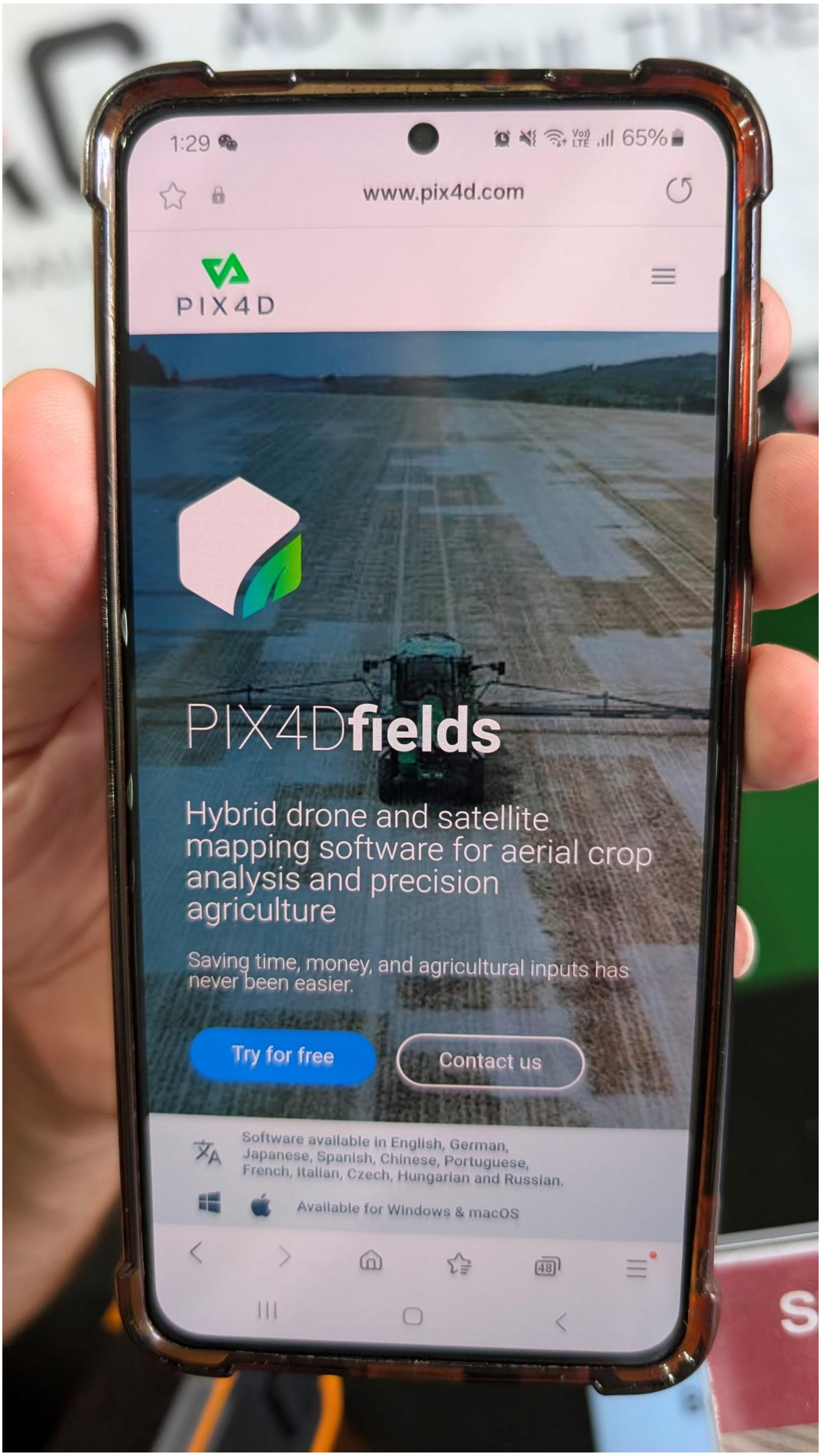
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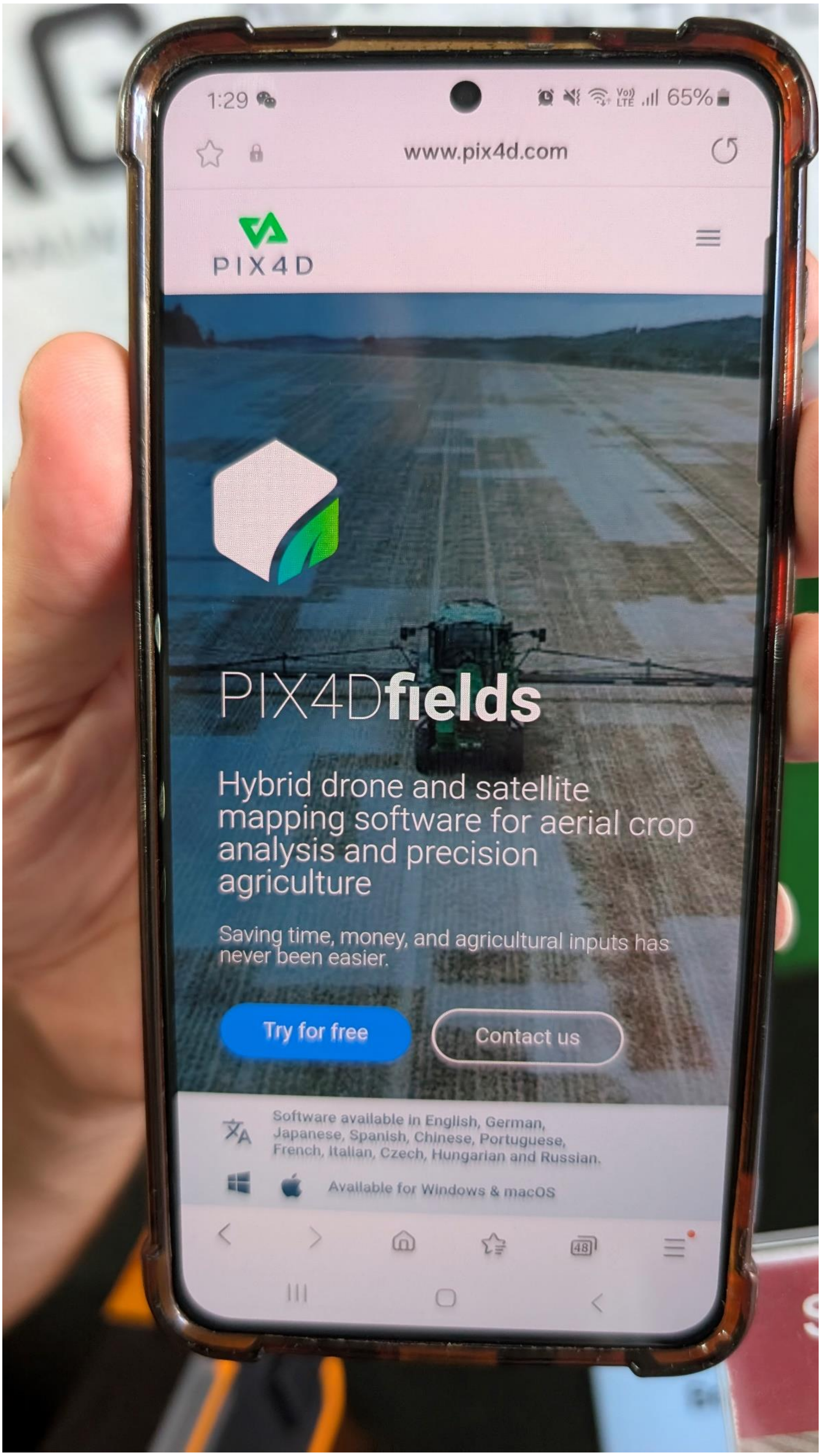
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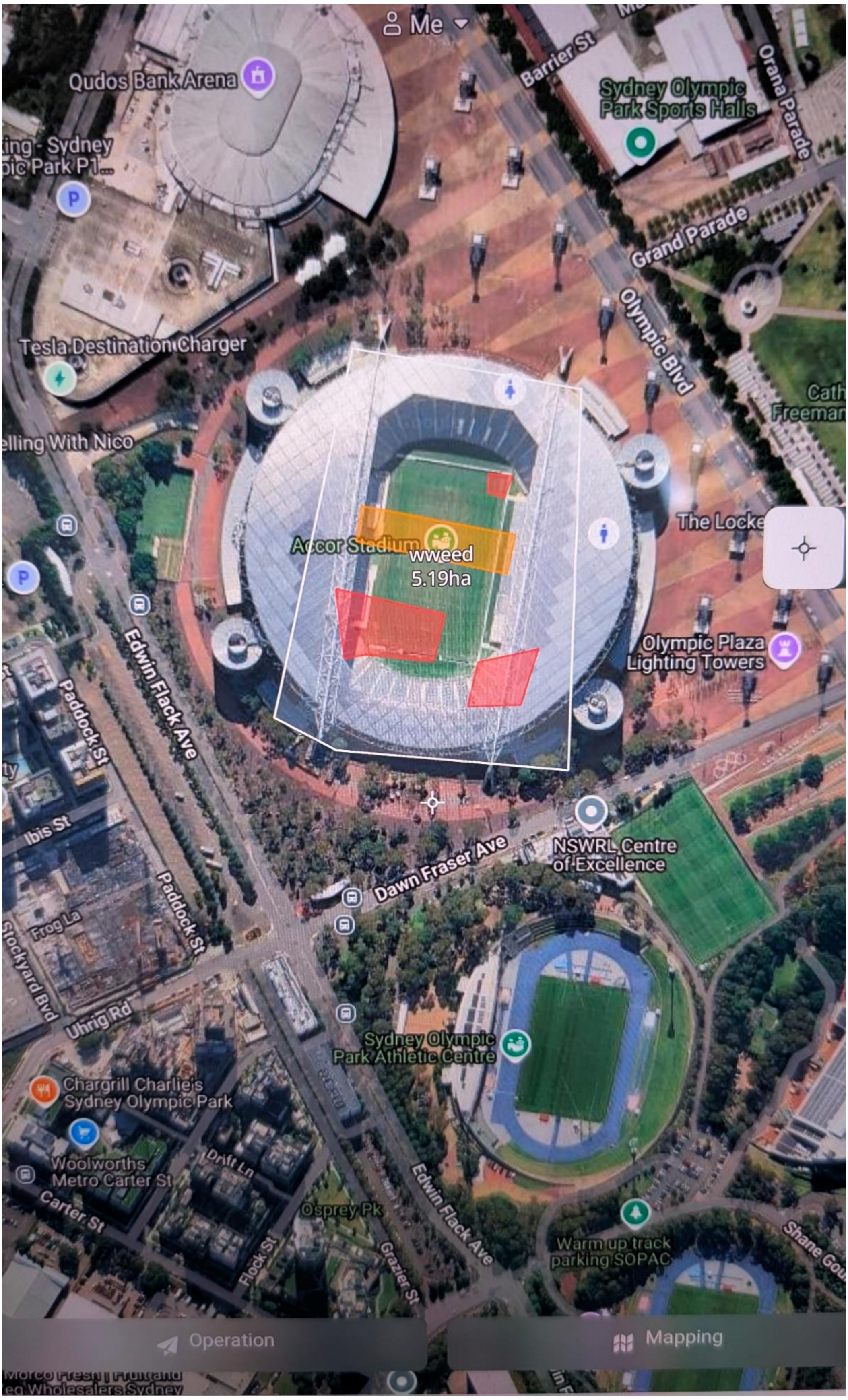
Available for Windows & macOS
48



This is how the App looks… Red is no go area and orange is spray area